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Gyroscopic Stability and Attitude Control During Atmospheric Ascent

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1. Gyroscopic Stability Fundamentals

A spinning body resists angular perturbation via gyroscopic stiffness. The precession equation:

$$\Omega_{\text{precession}} = \tau / L_{\text{gyro}}$$

Where τ = applied torque, $L_{\text{gyro}} = I_{\text{axial}} * \omega_{\text{spin}}$ (angular momentum).

1.1 Required Spin Rate for Stability

Condition for gyroscopic stability: precession rate $\ll$ angular velocity of trajectory change

For the AMSR-UBSCLS spear, the dominant perturbing torque is aerodynamic:

$$\tau_{\text{aero}} = 0.5 * \rho * v^2 * A * D * C_{m_alpha} * \alpha$$

Where $C_{m_alpha} = 0.2/\text{rad}$ (aerodynamic pitching moment coefficient for finned cylinder, empirical), and α = angle of attack (max 2° in turbulent launch environment).

At worst case (Mach 3, $h=15$ km): $\rho=0.195$ kg/m³, $v=900$ m/s, $A=0.00503$ m², $D=0.08$ m, $\alpha=0.035$ rad:

$$\tau_{\text{aero}} = 0.5 * 0.195 * 810000 * 0.00503 * 0.08 * 0.2 * 0.035 = 1.11 \text{ N}\cdot\text{m}$$

Required ω_{spin} for precession rate < 1 deg/s:

$$\omega_{\text{min}} = \tau / (I_{\text{axial}} * 0.0175) = 1.11 / (3.8\text{e-}4 * 0.0175) = 167 \text{ rad/s} = 1596 \text{ RPM}$$

Required spin rate: > 1600 RPM. Our 2000 RPM exit spin (from Vol 01) provides adequate margin.

1.2 Spin Decay During Atmospheric Ascent

Air friction decays spin rate. From aerodynamic torque on spinning body:

$$\tau_{\text{spin_decay}} = 0.5 * \rho * \omega * v * R^3 * C_{m_q} * \pi$$

Numerically integrating from surface to 100 km: - Surface (dense air, high drag): ω decreases from 2000 RPM to ~ 1600 RPM in first 5 km - Above 15 km: negligible spin decay (air density too low) - At 100 km: ω still ~ 1580 RPM — **adequate throughout trajectory**

2. Fin Stabilization — Passive Backup

Even with gyroscopic stabilization, passive aerodynamic stability is beneficial.

2.1 Fin Design

Four cruciform fins at spear base, from same ZrB₂-SiC as body: - Fin span: $s = 0.5 * D = 0.04$ m - Fin chord: $c = 0.3 * L = 0.192$ m - Fin planform area per fin: $S_{\text{fin}} = 0.04 * 0.192 = 0.00768$ m² - Fin material: ZrB₂-SiC thin sheet, 2mm thick

Static margin analysis: Center of pressure (C_p) location (finned rocket formula, Barrowman method):

$$X_{\text{cp}} = X_{\text{body_cp}} + 4 * N_{\text{fin}} * S_{\text{fin}} * (X_{\text{fin_mid}} - X_{\text{cp_body}}) / (A_{\text{body}} * C_{n_alpha_body})$$

Simplified: $X_{cp} = 0.65 * L$ from nose (finned configuration) Center of gravity: $X_{cg} = 0.45 * L$ from nose (front-heavy: screw + SRM forward)

Static margin = $(X_{cp} - X_{cg}) / D = (0.65 - 0.45) * L / D = 0.20 * 0.64 / 0.08 = 1.6 \text{ calibers}$

Static margin 1.6 cal > 1.0 cal minimum: STABLE.

2.2 Fin Thermal Analysis

At Mach 8+ (Scale 4-5), fin leading edges experience severe heating:

$T_{\text{leading_edge}} = T_{\text{stag}} * (1 - 0.05 * M^2)^{-1} \quad T_{\text{stag}} * 1.4 \quad [\text{at } M=8]$

At Mach 8, $h=20 \text{ km}$: $T_{\text{stag}} = 4557 \text{ K} \rightarrow T_{\text{leading_edge}} = 6380 \text{ K}$

ZrB₂-SiC melting: $3245^\circ\text{C} = 3518 \text{ K}$ — insufficient!

Solution for Scale 4-5: Replace fin leading edges with HfB₂ tiles ($T_{\text{melt}}=3660 \text{ K}$): Still marginal at $M=8$. Use thermal ablation as protective mechanism: - HfB₂ ablation rate: 0.1 mm/s at 3600 K (1 mm ablated per 10 seconds at max heating) - For 5-second maximum high-speed phase: 0.5 mm ablated — acceptable single-use design

3. Attitude Control Active System (Scale 4-5 Only)

For orbital-class payloads, passive fins may not provide sufficient control. Active attitude control using KNbO₃-BaTiO₃ piezo-driven control vanes:

Vane deflection: $\pm 5^\circ$ provides $\sim 120 \text{ N}$ sideforce at Mach 4 Required torque for $1^\circ/\text{s}$ pitch rate on 100 kg spear: $\tau = I_{\text{transverse}} * \alpha_{\text{dot}} = 0.8 * 0.017 = 0.014 \text{ N} \cdot \text{m}$ (tiny)

Conclusion: KNbO₃-BaTiO₃ vane actuation is more than adequate for attitude control.

Citations (Vol 19)

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